

Information Retrieval (IR) is the computer science discipline focused on searching for documents and relevant passages in large unstructured collections. Traditional IR systems rely on keyword indexing, inverted indices, and lexical term matching rather than neural language models.

Key Classical IR Algorithms:

1. TF-IDF (Term Frequency-Inverse Document Frequency): A statistical measure evaluating how relevant a word is to a document in a collection. It scores documents based on word frequency adjusted by corpus rarity.
2. BM25 (Best Matching 25): A probabilistic ranking function widely used in search engines to rank matching documents according to query term occurrences, incorporating document length normalization.

Classical search engines excel at exact keyword precision and high-throughput lookups, but they cannot natively resolve semantic synonyms, paraphrase variations, or conceptual meaning without exact term overlap.

Relational Database Management Systems (RDBMS) organize data into structured tables consisting of rows and columns governed by formal schemas. Data access and manipulation are executed using Structured Query Language (SQL), grounded in relational algebra and first-order predicate logic.

Core Database Principles:

1. ACID Guarantees: Relational database engines guarantee Atomicity, Consistency, Isolation, and Durability to ensure strictly deterministic, reliable transaction execution.
2. Deterministic Query Execution: Unlike generative artificial intelligence or probabilistic machine learning models, relational database query engines return exact, reproducible data subsets without hallucination or approximation.

Modern data architectures frequently combine deterministic relational databases with semantic vector indices, uniting structured integrity with unstructured contextual search capabilities.